



PRODUCT
DATASHEET '24

ADVANCE ANALYTIK

REVOLUTIONIZING ONLINE MONITORING SOLUTIONS



OPTICS 1000 SERIES

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Optic 1000 Series

The Optic 1000 Series is a high-performance online water analyser that provides accurate and reliable measurements for a wide range of water quality parameters such as Ammonium, Nitrates, Nitrites, Total Nitrogen, Phosphates, Total Phosphates, Organic Matter, Total Organic Carbon, Sulfides, Sulfites, Total Sulfur, Total Suspended Solids, Turbidity, Color, Chlorine, Chlorine Dioxide, Hydrocarbon, Aromatic Compounds, Phenol, Dissolved Oxygen, Conductivity, Ozone and pH. Designed with state-of-the-art optical technology, it ensures exceptional stability, low operating costs, and high precision for various industries, including wastewater treatment plants and river monitoring stations. This device offers versatility for a wide range of applications also, the user-friendly interface enables user to pick any 4 modules and any 8 parameters of choice to be added in this modular analyser.



Key Features

- **Multi-parameter monitoring:** Simultaneous measurement of organic matter, nitrate, color, turbidity, and more.
- **Advanced Optics:** High-resolution optics with a scanning wavelength range from 180 to 750 nm.
- **Durability:** Built-in automatic cleaning system with long-lasting lamps.
- **Modular Design:** Expandable with external probes for additional measurements like pH, conductivity, and dissolved oxygen.
- **User-Friendly Interface:** Large, intuitive touch screen for easy operation and maintenance diagnostics.



Our analyzers come with a 1-year parts warranty, provided they are used under normal operating conditions.

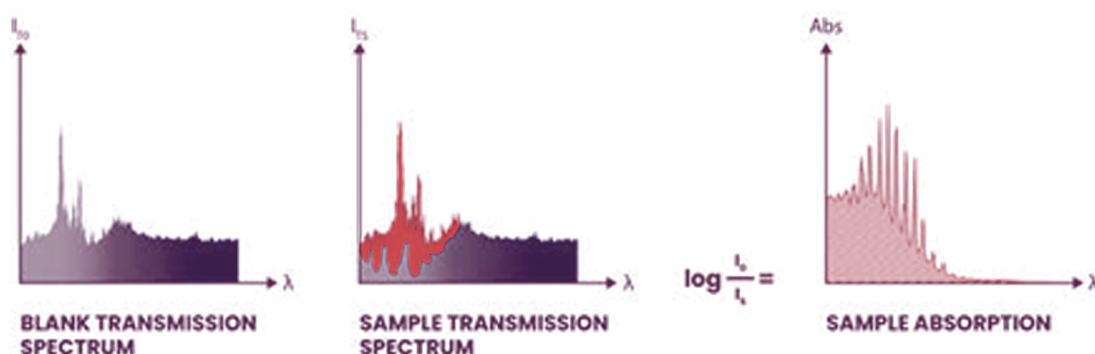


OUR MEASUREMENT TECHNOLOGY

UV VISIBLE SPECTROSCOPY

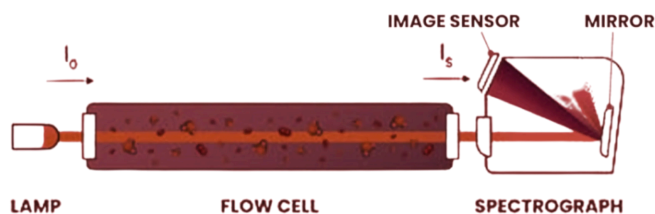
By leveraging a high-resolution spectrograph combined with advanced mathematical processing, this method achieves heightened precision and selectivity, enabling us to attain superior levels of accuracy and specificity.

The measurement principle relies on UV absorption spectrum according to Beer-Lambert's Law, where the calculation of the absorption spectrum involves determining the difference between the incident light (I_0) received on the reference (zero) and the transmitted light (I_t) passing through the sample.

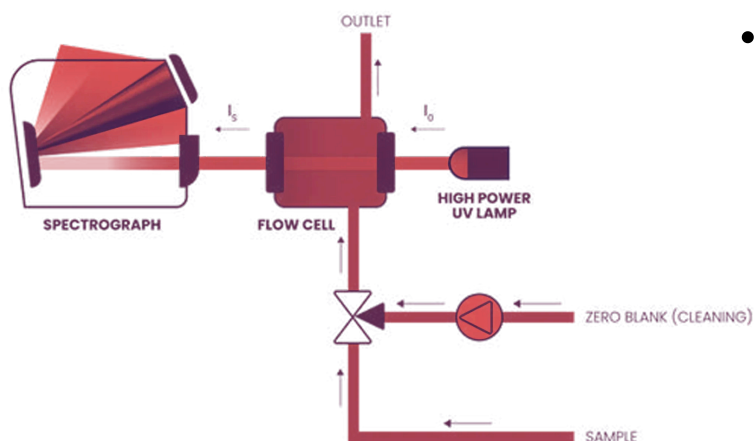


Absorbance (A) is determined using the formula: $A = \log(I_0 / I_t)$

The relationship between the concentration of molecules (c), the absorbance spectrum (A), and the optical path (l) of the measuring cell is linear.



The absorption coefficient (ϵ) is defined by the formula: $\epsilon = A / (lc)$.



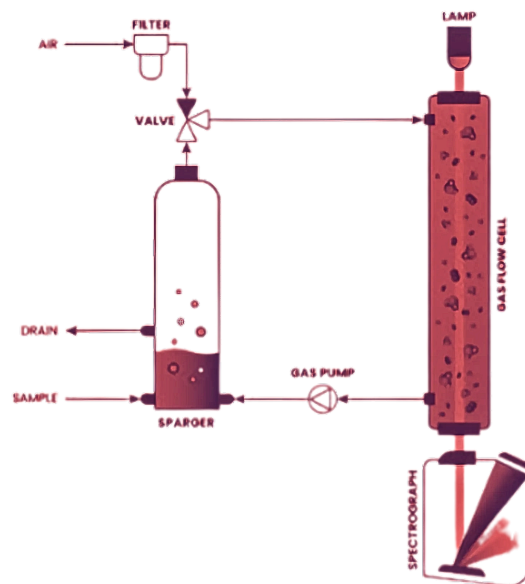
• Liquid Phase Measurement

For liquid phase measurement, the neutral fluid is injected into the measurement cell to obtain a baseline (blank). The sample is then analyzed by injecting it into the measurement cell, where its absorption spectrum is captured. Mathematical processing is applied to derive the concentration of the target compound.

UV VISIBLE SPECTROSCOPY

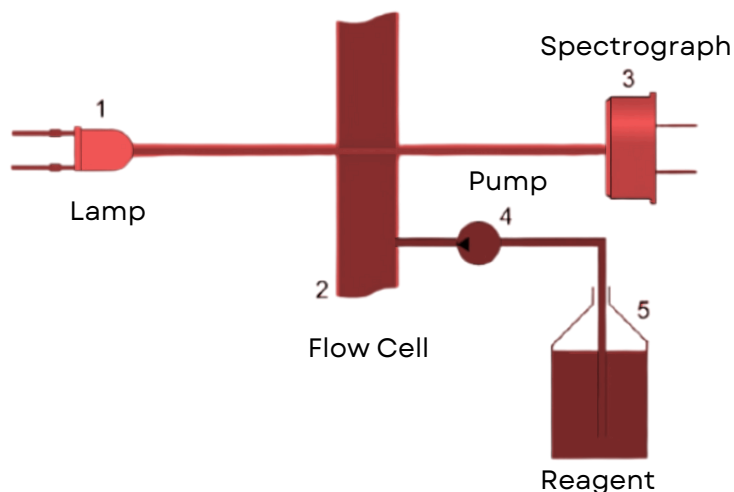
• Gas Phase Measurement

In gas phase measurement, the liquid sample is first injected into glass sparger where reagent gets added and then due to exothermic reaction the entire residual molecule gets converted into gaseous phase which further gets carried to the gas flow cell using ambient air. Afterward, the sample is introduced into the cell the absorption spectrum is captured using UV spectroscopy and the concentration of the compound is determined through FFT mathematical treatment.



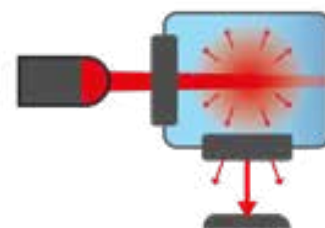
COLORIMETRY

It is based on a color-developing chemical reaction carried out inside the analyser flow cell. Sample is injected by the sample pump and mixed with the appropriate reagents to enable a reaction. The analyser uses Xenon Lamp and a photodiode to photometrically measure the developed color intensity (absorbance). The absorbance is proportional to the concentration of the sample (analyte).



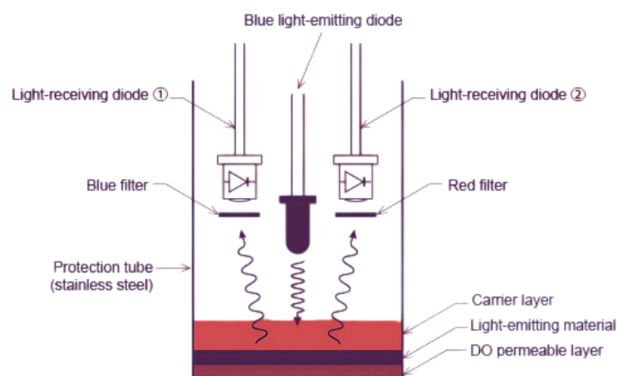
IR NEPHELOMETRY

IR (infrared) nephelometry is a technique used to measure Total Suspended Solids (TSS) in a liquid medium, representing the content of suspended matter that disrupts the fluid. This measurement adheres to the ISO 7027 / EN 27027 standard. During the process, an IR light beam traverses the sample and scatters in various directions. A detector then transforms the scattered light's intensity into an electrical signal. The turbidity value increases with higher light intensity.



VISIBLE FLUORESCENCE

This method uses the quenching phenomenon of oxygen to calculate the level of DO. Fluorescence from a fluorescent material excited by light from a blue light-emitting diode (LED) is quenched by oxygen passed through a DO permeable layer. The higher the DO level is, the stronger the quenching phenomenon is and the less fluorescence the detector (light-receiving diodes) detects.



POTENTIOMETRY

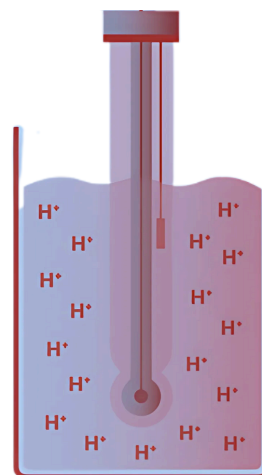
The pH of a solution is determined as the negative logarithm of the molar concentration of hydronium ions (H_3O^+). The voltage between the reference and sensing half-cells is governed by the Nernst equation:

$$V = V_0 + \left(\frac{RT}{nF} \right) \ln \left(\frac{[\text{H}_m^+]}{[\text{H}_{ref}^+]} \right)$$

Where :

- V_0 : Voltage of the reference half-cell
- R: Ideal gas constant
- T: Absolute temperature
- n: Valence
- F: Faraday constant
- $[\text{H}_m^+]$: Concentration of H^+ ions in the sample
- $[\text{H}_{ref}^+]$: Reference H^+ concentration

Absorbance (A) is given by the equation: $A = \log(I_0/I)$.

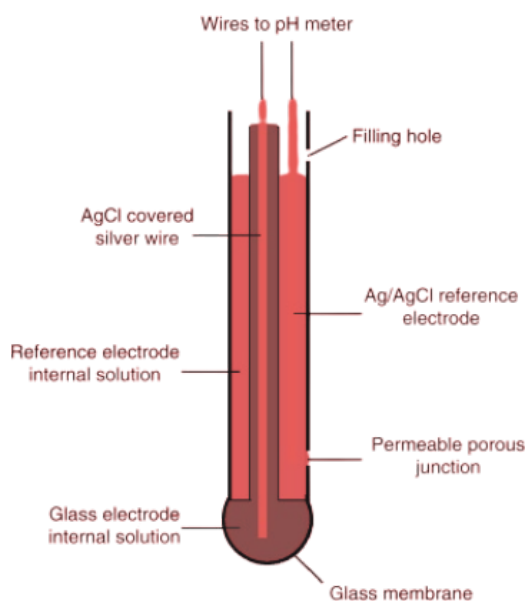


The measurement method is based on the potential difference between two half-cells (reference and sensing). These half-cells consist of a conductor immersed in an appropriate electrolyte solution, separated by a conductive glass membrane. Typically, both half-cells are housed within a single electrode.

At 25°C, one pH unit corresponds to a potential difference of 59.16 mV. For pH 7, the output is 0V at all temperatures.

ISE

Ion-selective membranes selectively interact with target ions through their special chemical properties. This interaction causes a change in the concentration of ions on the membrane, leading to a corresponding change in the electrolyte within the electrode body, resulting in a change in potential. When the target ion reacts with the ion-selective membrane, the potential inside the electrode body changes and this change can be measured indirectly by measuring the potential of the electrode to measure the concentration of the target ion. Thus, an ion-selective electrode can determine the concentration of a specific ion in solution by measuring the electrode's potential change



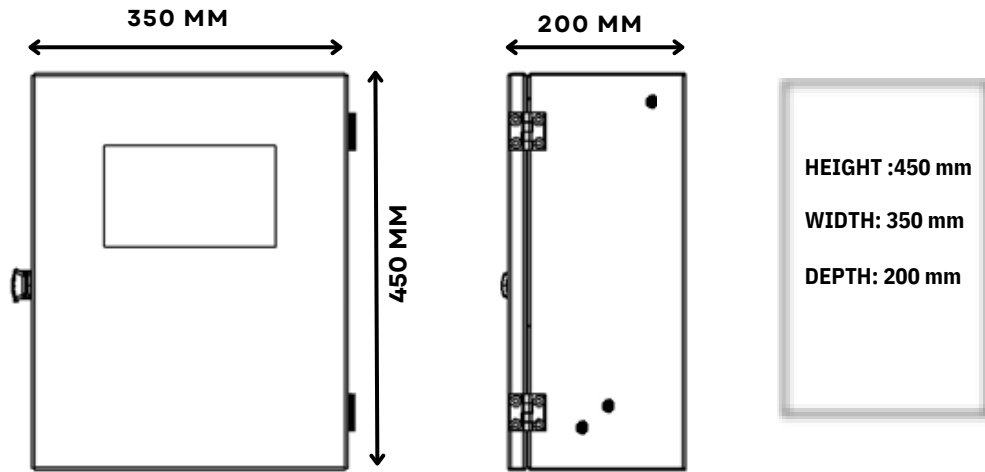
Parameter Specification of Optic 1000 Series

UV SPECTROSCOPY	
Hydrocarbon (Oil) – OIW	0.1 ... 100 mg/L*
Nitrates – NO ₃ ⁻	0.1 ... 100 mg/L*
Nitrites – NO ₂ ⁻	0.1 ... 10 mg/L*
Total Nitrogen – TN	0.1 ... 100 mg/L*
UV SPECTROSCOPY - GAS PHASE	
Ammonium – NH ₄ ⁺	0.1 ... 1000 mg/L*
Sulfides – S ₂ ⁻	0.1 ... 100 mg/L*
Sulfites – SO ₃ ²⁻	0.1 ... 100 mg/L*
Total Sulfur – TS	0.1 ... 100 mg/L*
Chlorine – Cl ₂	0.1 ... 10 mg/L*
Chlorine Dioxide – ClO ₂	0.1 ... 10 mg/L*
Total Chlorine – TC	0.1 ... 10 mg/L*
Aromatic Compounds – BTEX	0.1 ... 100 mg/L*
Phenol – C ₆ H ₅ OH	0.1 ... 100 mg/L*
Ozone – O ₃	0.1 ... 10 mg/L*
COLORIMETRY	
Phosphates – PO ₄ ³⁻	0.1 ... 10 mg/L*
Total Phosphates – TP	0.1 ... 10 mg/L*
IR - NEPHELOMETRY	
Total Suspended Solids – TSS	0.1 ... 10000 mg/L*
Turbidity – Turb	0.1 ... 100 NTU*
IR SPECTROSCOPY - GAS PHASE	
Total Organic Carbon – TOC	0.1 ... 10000 mg/L*
VISIBLE SPECTROSCOPY	
Color	0.1 ... 1000 Pt-Co*
VISIBLE FLUORESCENCE	
Dissolved Oxygen – DO	0.1 ... 100 mg/L*
CONDUCTIMETER	
Conductivity – EC	0.1 ... 1000 mS/cm*
POTENTIOMETRY	
pH – H ⁺	0 ... 14*

*Note - Other ranges are available on request.



TECHNICAL SPECIFICATION



ENCLOSURE	
Type	Panel Mounted/Wall Mounted
Material	SS316L
Dimensions (HxWxD)	450x350x200 mm
Weight	± 15 kg
Protection class	IP65
Area Classification	Safe, Zone 1 or 2 <i>(Available on request)</i>
SAMPLE CONDITION	
Flow	0.2 L/min
Pressure	< 2 bars
Temperature	0... 60°C
CONTROLLER	
Display	8.5" TFT colour screen 16/9 (LED backlight)
Resolution	800 x 480 px
Touch screen	Glass to glass
Memory	16 GB SD card
Data transfer	USB type A
Operating temperature	0... 50°C
Operating humidity	< 90 % RH
COMMUNICATION OUTPUT	
Analog	4-20 mA isolated (Active or Passive) / 500 Ω max
Relay	Programmable limit or fault alarms / 5A (NO) 3A (NC) @ 277 VAC
Digital	RS485 / Modbus (Slave or Master) - Show value, calibration status, fault status...
POWER SUPPLY	
Voltage	100 ... 240 VAC (50 - 60 Hz) or 24 VDC (Universal)
Consumption	< 20 W (60 W max.)

Sensors Compatible with Optics 1000 Series

ORDERING CODE	DESCRIPTION
VizSens-TurbLo	Low Turbidity Sensor Range: 0.001 - 100 NTU Cable Length - Standard 3m (Maximum can be extended to 30m)
VizSens-NO3N	Nitrate Nitrogen Sensor Range: 0-3000mg/L NO3N Cable Length - Standard 5m (Maximum can be extended to 100m)
VizSens-NH4N	Ammonia Nitrogen Sensor Range: 0.1-3000mg/L NH4N Cable Length - Standard 5m (Maximum can be extended to 100m)
VizSens-OIW	Oil in Water Sensor Range: 0-50ppm Cable Length - Standard 10m (Maximum can be extended to 50m)
VizSens-BGA	Blue Green Algae Sensor Range: 0-300000 cells/mL Cable Length - Standard 10m (Maximum can be extended to 100m)
VizSens-CPHL	Chlorophyll Sensor Range: 0-500 ug/L Cable Length - Standard 10m (Maximum can be extended to 100m)
VizSens-Turb	Optical Turbidity Sensor Range: 0.01 - 4000 NTU Cable Length - Standard 10m (Maximum can be extended to 100m)
VizSens-SS	Suspended Solids/Sludge Sensor Range: 0.01-20000 mg/L Cable Length - Standard 10m (Maximum can be extended to 100m)
VizSens-RCL	Residual Chlorine Sensor Range: 0-20mg/L Cable Length - Standard 10m (Maximum can be extended to 100m)
VizSens-ODO	Optical DO Sensor Range: 0-20 mg/L or 0-200% saturation Cable Length - Standard 10m (Maximum can be extended to 100m)
VizSens-EC (Digital)	Digital Conductivity Sensor Range: 0-20 mS/cm Cable Length - Standard 10m (Maximum can be extended to 200m)
VizSens-EC (Analog)	Analog Conductivity Sensor Range: 0-20 mS/cm Cable Length - Standard 10m (Maximum can be extended to 20m)
VizSens-pH (Digital)	Digital pH Sensor Range : 0-14 pH Cable Length - Standard 10m (Maximum can be extended to 200m)



ORDERING CODE	DESCRIPTION
VizSens-pH (Analog)	Analog pH Sensor Range : 0-14 pH Cable Length - Standard 10m (Maximum can be extended to 20m)
VizSens-ORP (Analog)	Analog ORP Sensor Range: -/+ 2000 mV Cable Length - Standard 10m (Maximum can be extended to 20m)
VizSens-TDS (Analog)	Analog TDS Sensor Range: 0-20000 ppm Cable Length - Standard 10m (Maximum can be extended to 20m)
VizSens-ISE-NH3	Ammonia (NH3) Range: 0.01 - 17,000 ppm as NH3
VizSens-ISE-CN-	Cyanide (CN-) Range: 0.2 - 260 ppm
VizSens-ISE-F-	Fluoride (F-) Range : 0.02 - 19000 ppm
VizSens-ISE-Na+	Sodium (Na+) Range : 0.01 - 100,000 ppm
VizSens-ISE-Cl-	Chloride (Cl-) Range : 1.8 - 35,000 ppm
VizSens-ISE-Ca2+/Mg2+	Water Hardness (Ca2+ / Mg2+) Range : 0.5 - 40,000 ppm





LET'S CONNECT !

Let's work together to find a solution that works for you



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